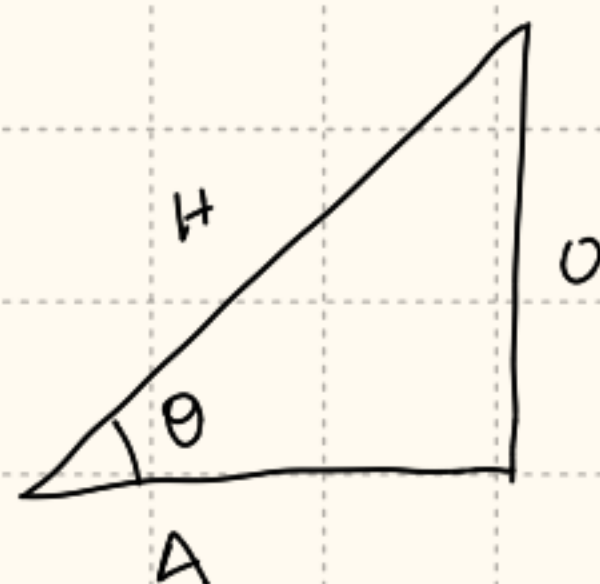


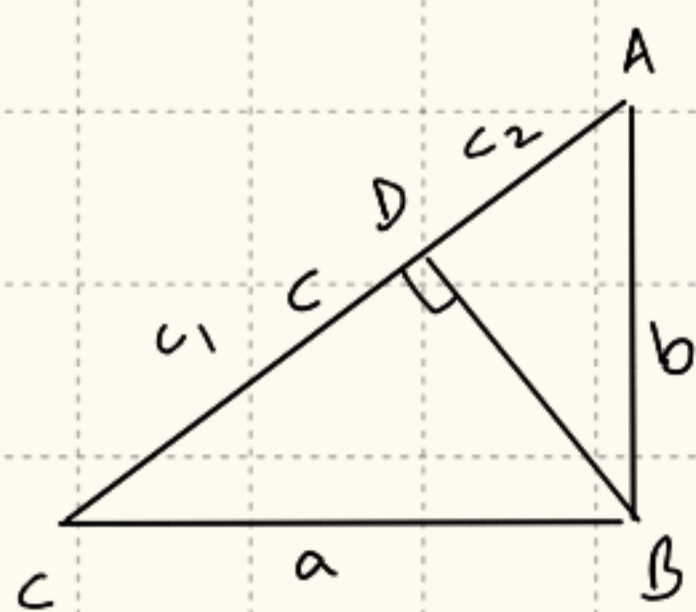
Trigonometry

$$\sin(\theta) = \frac{O}{H}$$

$$\cos(\theta) = \frac{A}{H}$$



Pythagoras Theorem



$\triangle ABC$ & $\triangle ABD$

$$\begin{aligned} AB &= AB \\ \angle DAB &= \angle CAB \\ \angle D &= \angle B = 90^\circ \\ \triangle DAB &\sim \triangle BAC \\ \frac{c_2}{b} &= \frac{b}{c} \quad (1) \\ b^2 &= c_2 c \end{aligned}$$

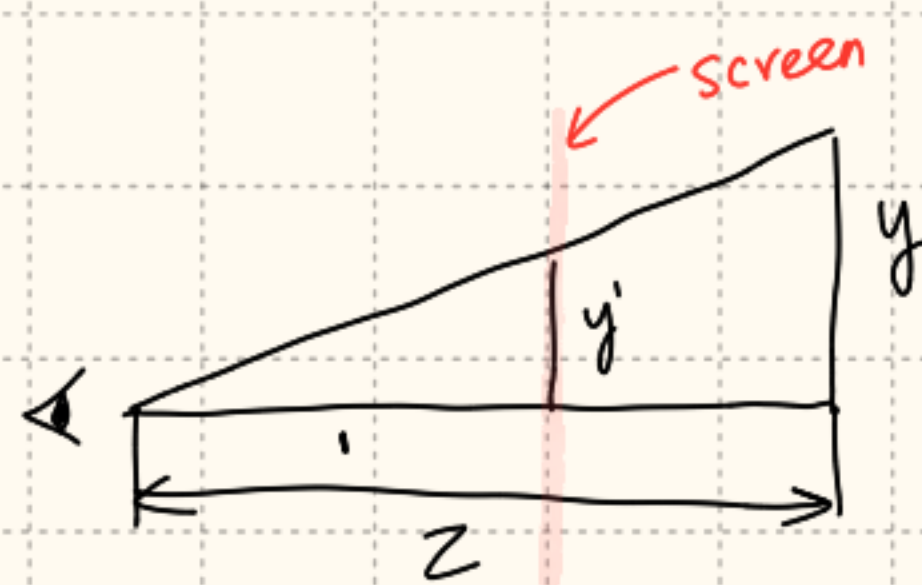
$\triangle BCD$ & $\triangle ABC$

$$\begin{aligned} BC &= BC \\ \angle DCB &= \angle ACB \\ \angle D &= \angle B = 90^\circ \\ \triangle DCB &\sim \triangle BCA \\ \frac{c_1}{a} &= \frac{a}{c} \quad (2) \\ a^2 &= c_1 c \end{aligned}$$

(1) + (2)

$$\begin{aligned} a^2 + b^2 &= c \cdot (c_1 + c_2) \\ a^2 + b^2 &= c^2 \end{aligned}$$

Projection on screen

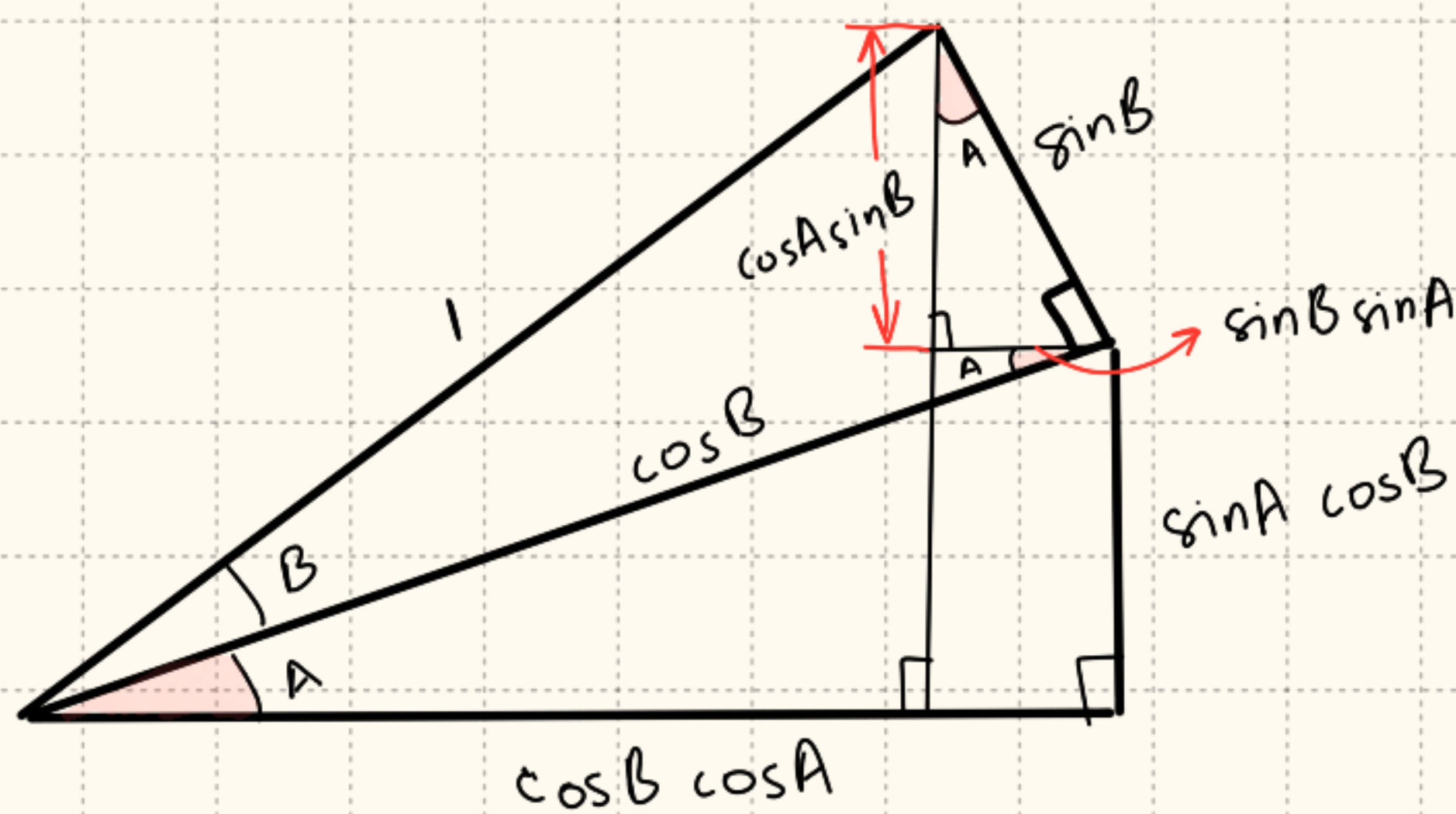


$$\frac{y'}{z'} = \frac{y}{z} \Rightarrow y' = \frac{y}{z} z'$$

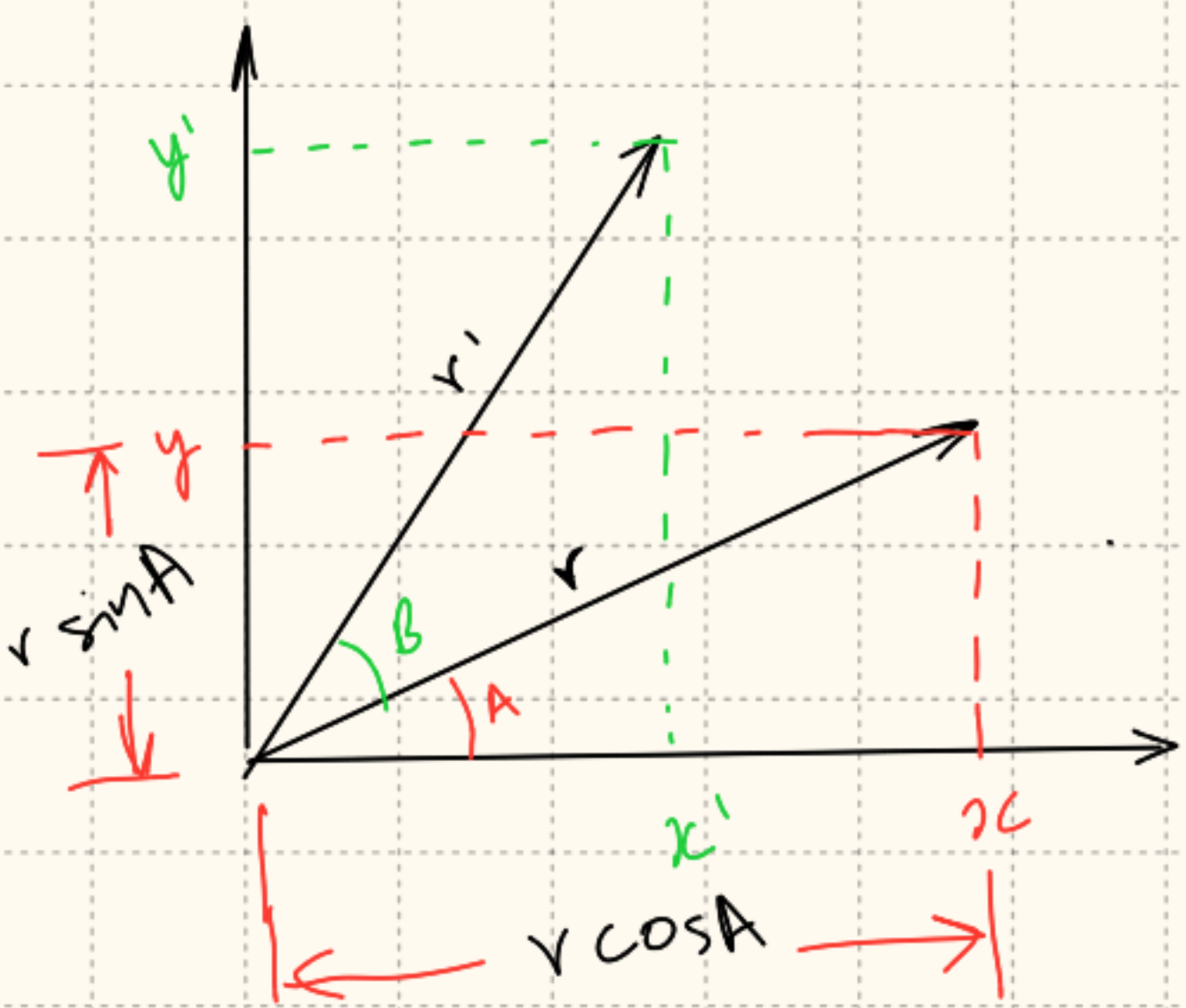
$$\text{Similarly } x' = \frac{x}{z} z'$$

Just assume that z' is 1

Rotation Matrix (The why of psi)



$$\begin{aligned} \sin(A+B) &= \sin A \cos B + \sin B \cos A \\ \cos(A+B) &= \cos A \cos B - \sin A \sin B \end{aligned}$$



You need this because the starting rotation should be accounted for

Consider the vector lying on x -axis. The y component will be zero

$|r| = |r'|$ magnitude is same

$$\begin{aligned} r' &= r \\ x' &= r \cos(A+B) \\ x' &= r(\cos A \cos B - \sin A \sin B) \Rightarrow x \cdot r \cos B - y \cdot r \sin B \\ y' &= r \sin(A+B) \\ y' &= r(\sin A \cos B + \sin B \cos A) \Rightarrow x \cdot r \sin B + y \cdot r \cos B \end{aligned}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos B & -\sin B \\ \sin B & \cos B \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{if } r=1$$